Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 1.0 | **Date:** 04-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Phase II/III Multicenter Study Evaluating the Efficacy and Safety of Multiple Targeted Therapies as Treatments for Patients With Advanced or Metastatic Non-Small Cell Lung Cancer (NSCLC) Harboring Actionable Somatic Mutations Detected in Blood (B-FAST: Blood-First Assay Screening Trial) **Short Title/ Acronym:** A Study to Evaluate the Efficacy and Safety of Multiple Targeted Therapies as Treatments for Participants With Non-Small Cell Lung Cancer (NSCLC) / B-FAST Study **Protocol Number:** BO29554 **NCT Number:** NCT03178552 **Sponsor Name:** Roche (Hoffmann-La Roche) **Investigational Product:** Multiple targeted therapies and immunotherapies including Alectinib, Atezolizumab, Pemetrexed, Cisplatin (e.g., Platinol), Carboplatin, Gemcitabine, Entrectinib, Cobimetinib, Vemurafenib, Bevacizumab, Divarasib, Docetaxel **Phase of Development:** Phase II/III **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and efficacy of targeted therapies or immunotherapy as single agents or in combination in participants with unresectable, advanced or metastatic NSCLC determined to harbor oncogenic somatic mutations or positive by tumor mutational burden (TMB) assay as identified by a blood-based next-generation sequencing (NGS) circulating tumor DNA (ctDNA) assay. **Secondary Objectives:** To assess Investigator-assessed clinical benefit rate (CBR), Duration of Response (DOR), Progression-Free Survival (PFS), Overall Survival (OS), Time to central nervous system (CNS) progression, and overall safety profile.

2.2 Background & Rationale To address the need for a less invasive diagnostic alternative with faster turnaround times to identify targetable alterations and provide appropriate biomarker-informed treatments for patients with advanced NSCLC. The study seeks to validate the clinical utility of liquid biopsies (blood-based next-generation sequencing [NGS] circulating tumor DNA assay) to inform first-line clinical decision-making without relying solely on tissue-based testing.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator for some cohorts (e.g., Atezolizumab vs. platinum-based chemotherapy) and Single-arm for others (e.g., ROS1 or ALK targeted cohorts) **Blinding:** Open-label **Randomization:** Randomized (e.g., 1:1 in Cohort C) or N/A depending on the specific biomarker cohort

3.2 Planned Enrollment & Duration Trial Status: Active, not recruiting (ACTIVE_NOT_RECRUITING) **Target N:** 1,000 evaluable patients **Number of Sites:** Global, multicenter **Estimated Study Start:** 22-Sep-2017 **Estimated Primary Completion:** 31-Oct-2026

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically or cytologically confirmed diagnosis of unresectable Stage IIIb not amenable to treatment with combined modality chemoradiation (advanced) or Stage IV (metastatic) NSCLC. 2. Eastern Cooperative Oncology Group (ECOG) Performance Status 0-2. 3. Measurable disease. 4. Adequate recovery from most recent systemic or local treatment for cancer. 5. Adequate organ function. 6. Life expectancy ≥ 12 weeks. 7. For female participants of childbearing potential and male participants, willingness to use acceptable methods of contraception.

4.2 Exclusion Criteria 1. Inability to swallow oral medication. 2. Women who are pregnant or lactating. 3. Symptomatic, untreated CNS metastases. 4. History of malignancy other than NSCLC within 5 years prior to screening with the exception of malignancies with negligible risk of metastasis or death. 5. Significant cardiovascular disease, such as New York Heart Association cardiac disease (Class II or greater), myocardial infarction, or cerebrovascular accident within 3 months prior to randomization, unstable arrhythmias, or unstable angina. 6. Known active or uncontrolled human immunodeficiency virus (HIV) infection. 7. Either a concurrent condition or history of a prior condition that places the patient at unacceptable risk if he/she were treated with the study drug or confounds the ability to interpret data from the study. 8. Inability to comply with other requirements of the protocol.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Varies per assigned cohort based on mutation (e.g., Entrectinib 600 mg orally once daily for ROS1+; Alectinib 600 mg twice daily for ALK+; Atezolizumab infusions for bTMB cohorts). **Route of Administration:** Oral or Intravenous infusion (depending on the assigned agent). **Duration of Treatment:** Administered until disease progression, unacceptable toxicity, withdrawal of consent, or study termination.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care measures required for co-morbidities. **Prohibited:**

Prior treatment with systemic therapies for advanced/metastatic NSCLC; concurrent investigational agents or live vaccines; robust CYP3A/CYP2D6 inhibitors or inducers (varies by cohort protocol).

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6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, Blood-based NGS (ctDNA assay) for biomarker allocation | | **Baseline** | Day 0 | Treatment assignment, First Dose, Safety baseline | | **Interim** | Every 3-4 Weeks | Study drug administration, Routine Safety Labs, AE monitoring | | **Tumor Assessment** | Every 8 Weeks | Radiographic tumor assessments (RECIST v1.1), Efficacy evaluation | | **End of Study** | Variable | Treatment Discontinuation visit, Survival follow-up |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: * **Cohort A:** Percentage of Participants with Confirmed Objective Response as Assessed by the Investigator Based on the Response Evaluation Criteria in Solid Tumors (RECIST) Version (v) 1.1. * **Cohort B:** Percentage of Participants with Confirmed Objective Response as Assessed by the Investigator Based on RECIST v1.1. * **Cohort C:** Progression Free Survival (PFS) as Assessed by the Investigator Based on RECIST v1.1 in bTMB PP1. **Measurement Method:** RECIST v1.1 evaluation by the investigator and corroborated by an Independent Review Facility (IRF).

7.2 Secondary & Exploratory Endpoints * **Cohorts A-F:** Duration of Response (DOR) as Assessed by the Investigator Based on RECIST v1.1. * **Cohorts A, B and D:** Percentage of Participants with Clinical Benefit Response as Assessed by the Investigator Based on RECIST v1.1. * **Cohorts A, B, D, F:** PFS as Assessed by the Investigator Based on RECIST v1.1. * Overall Survival (OS) * Time to central nervous system (CNS) progression

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring at regular study visits via clinical assessment, laboratory testing, and patient interviews. **Serious Adverse Events (SAEs):** Requires expedited reporting within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee ensures ongoing safety oversight across the multiple global cohorts.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy assessments; Safety Set for all participants receiving at least one dose of study medication. **Sample Size Justification:** Targeted total of approximately 1,000 patients, dynamically powered and structured across individual cohorts (A, B, C, D, etc.) to detect clinically meaningful improvements in ORR or PFS per actionable mutation prevalence. **Handling of Missing Data:** Standard oncology conventions including right-censoring for time-to-event endpoints (PFS, OS) and independent imputation techniques where defined in the cohort-specific SAP.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and centralized remote monitoring for source data verification. **Audit Trail:** Stringent validation via eCRF locking, tracking of ctDNA assay performance against standard tissue assays, and internal data clarification mechanisms.

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11. Study Identifiers & Governance

Posting ID: 992015398509459613 **Primary ID:** BO29554 **Registry Number:** NCT03178552 **Sponsor/Lead Organization:** Roche / Hoffmann-La Roche **Study Title:** B-FAST **Official Regulatory Title:** A Phase II/III Multicenter Study Evaluating the Efficacy and Safety of Multiple Targeted Therapies as Treatments for Patients With Advanced or Metastatic Non-Small Cell Lung Cancer (NSCLC) Harboring Actionable Somatic Mutations Detected in Blood (B-FAST: Blood-First Assay Screening Trial)

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12. Clinical Framework (Oncology/NSCLC Specific)

Note: Adapted from template to reflect the NSCLC focus of the requested NCT. **Primary Condition:** Non-Small Cell Lung Cancer (NSCLC) **Diagnosis Threshold:** Confirmed Stage IIIb/IV NSCLC harboring actionable somatic mutations (e.g., ALK, ROS1, RET, BRAF) or high blood-based Tumor Mutational Burden (bTMB ≥ 16) detected via liquid biopsy. **Medical Methodology:** Biomarker-informed Targeted Therapy and Immunotherapy **Optimization Target:** Maximal tumor reduction (Objective Response) and prolongation of Progression-Free Survival (PFS)

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13. Study Procedures & Methodology

Management Pathway: Blood-First Assay Screening (liquid biopsy/ ctDNA) for rapid biomarker assignment **Education Protocol (HCP):** Comprehensive training on liquid biopsy acquisition, targeted agent dosing protocols, and adverse event management. **Education Protocol (Patient):** Informed consent mapping, genomic profiling implications, and adherence training for oral targeted inhibitors. **Quality Audit Mechanism:** Investigator vs. Independent Review Facility (IRF) scan concordance tracking, and assay cross-validation (e.g., bTMB vs. tissue assays).

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14. Observation & Follow-Up Schedule

Total Duration: Continues until disease progression or unacceptable toxicity. **Onsite Visit Cadence:** Dosing/Safety monitoring every 3 to 4 weeks (dependent on specific study drug). **Remote Monitoring Frequency:** Routine survival follow-up assessments post-progression. **Checklist Review Interval:** Tumor assessments conducted strictly every 8 weeks.

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15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | | Clinical Operations Lead | [Name] | ___ |** ***[DD-MMM-YYYY]*** **| |** ***Lead Statistician*** **|** ***[Name]*** **|** _____ | [DD-MMM-YYYY] |